

Rethinking Underground Space Utilization in Metro Stations: A Case Study of Shanghai's Jing'an Temple

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Summary

This study examines optimizing urban underground space (UUS) in metro stations by integrating Space Syntax and data-driven methods, using Shanghai Jing'an Temple Metro Station as a case study.

By analyzing accessibility, visibility, and human flow patterns, it highlights how strategic design adjustments can balance pedestrian flow, enhance user satisfaction, and boost commercial benefits. Strategies include redesigning pathways to improve spatial connectivity and prioritizing retail and food facilities placement in high-visibility areas to maximize commercial revenue. These insights offer both theoretical and practical guidance for improving UUS utilization, particularly in high-density urban cores.

KEYWORDS: Urban Underground Space, Space Syntax, Data-driven design, Metro Stations, Pedestrian Flow

1. Introduction

With the continuous advancement of global urbanization, cities are increasingly facing dual pressures of resource scarcity and the limited capacity of urban spaces. The dwindling availability of surface space has prompted cities to explore underground spaces as a new development avenue. Rational development and utilization of urban underground space (UUS) have become a core issue in urban planning (Harvey & Tulloch, 2006; Law, 2012). In this context, the significance of underground spaces in metro stations has grown. As hubs of transit, economic activity, and public services, the underground space of metro stations — particularly public and commercial areas — serves dual functions of facilitating transportation and providing urban amenities, thus playing a crucial role in modern urban development (Xu & Chen, 2021; van der Hoeven & van Nes, 2014). However, many existing metro station underground spaces exhibit imbalanced utilization. Some areas are excessively congested, while others remain underutilized. Moreover, challenges persist in balancing enhanced user experiences with economic benefits, making urban underground space design and planning a complex endeavor.

To address these challenges, this study takes Jing'an Temple Metro Station in Shanghai as a case study. By integrating Space Syntax analysis with data-driven methods, the research proposes a comprehensive framework for evaluating and optimizing the performance of urban underground space layouts. Specifically, this study quantifies the accessibility and flow potential of underground spaces using Space Syntax analysis, while assessing spatial usage efficiency and user satisfaction through field research, user surveys, and computer simulations. Based on these multi-faceted methods, this paper puts forward design optimization strategies aimed at providing theoretical support and practical references for the efficient utilization of urban underground spaces.

2. Methods

This study focuses on the urban underground spaces of Shanghai's Jing'an Temple Metro Station, which include Underground Public Space (UPS) and Underground Commercial Space (UCS). To investigate the structural characteristics of these spaces and their interactions with user behavior, the following methods were employed:



Figure 1 Shanghai's Jing'an Temple Metro Station

First, Space Syntax was applied to analyze the spatial properties of underground public and commercial areas. Specifically, Integration, Choice, and Visibility Graph Analysis (VGA) were used to quantify and evaluate spatial connectivity, central areas, and visibility. Integration analysis helped identify the most highly accessible areas within the underground space, while Choice analysis provided insights into critical nodes for pedestrian flow. Visibility Graph Analysis highlighted the potential influence of visual accessibility on user behavior.

Second, data-driven methods were employed to investigate the actual utilization of underground spaces, including: (1) field surveys and pedestrian flow measurements: monitoring key entrances and corridors to evaluate actual patterns of pedestrian movement and spatial vitality; (2) user experience surveys: utilizing a questionnaire designed to assess space accessibility, comfort, and usage efficiency to gather users' subjective feedback; and (3) agent-based behavior simulations: modeling passenger movement paths and dwell times under varied spatial layout scenarios to explore the potential impact of spatial adjustments on pedestrian flows and economic outcomes.



Figure 2 Entrance Traffic Survey

Finally, leveraging the insights from the above analyses, this study proposed specific spatial optimization strategies and tested their feasibility and efficacy through simulation-based validation.

3. Results

This research conducted a comprehensive analysis of the spatial performance and user behavior within underground spaces at Jing'an Temple Metro Station. The key findings are as follows:

First, Space Syntax analysis revealed that the Integration and Choice values of underground spaces significantly influence pedestrian flow efficiency and spatial vitality. High-Integration and high-Choice areas, such as the main station halls and primary corridors, constitute the core flow zones of underground spaces. These areas support both short-distance and long-distance pedestrian demands while functioning as critical hubs for traffic distribution. In contrast, areas with low Integration and Choice values are more likely to be overlooked or left underutilized. Thus, optimizing the structural relationship between primary and secondary corridors, as well as enhancing the layout of core pathways, can improve pedestrian flow efficiency and alleviate congestion in specific zones.

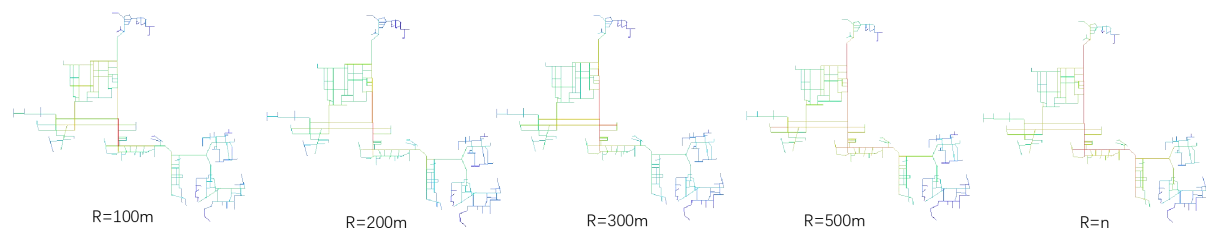


Figure 3 Segment Analysis-Integration

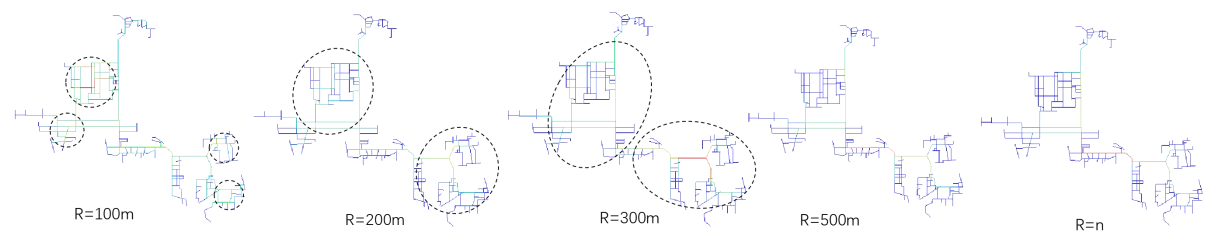


Figure 4 Segment Analysis-Choice

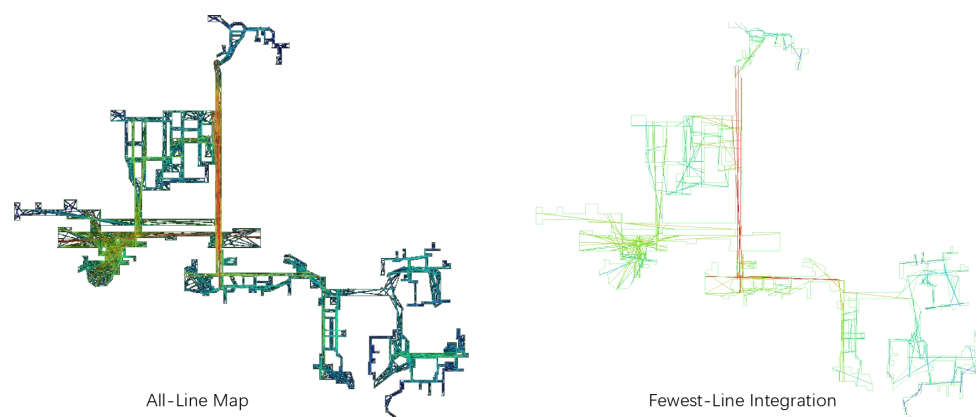


Figure 5 All-line Analysis

Second, the study demonstrated a strong correlation between spatial visibility and user experience. Visibility Graph Analysis indicated that highly visible areas, such as dining zones and node intersections, not only attract a higher proportion of passengers to dwell but also significantly enhance user satisfaction. The design of these areas should be further optimized by improving signage systems and increasing their visual appeal to heighten user engagement and spatial efficiency. Combining pedestrian flow measurements and path modeling, the study also suggested that improving the visibility of secondary commercial nodes could enhance commercial revenues and promote functional

synergy throughout the space.

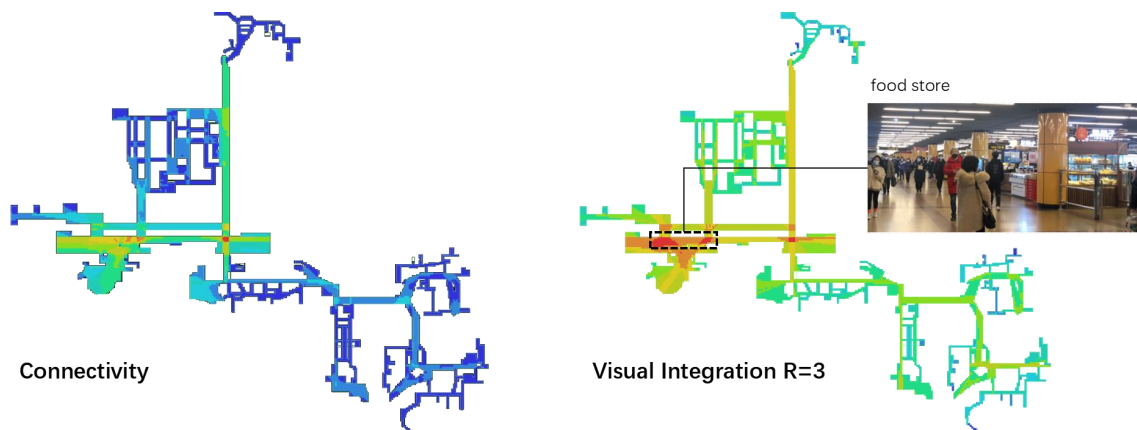


Figure 6 Visibility Graph Analysis

In addition, optimizing the layout of underground commercial spaces plays a crucial role in boosting economic efficiency and spatial usage. Agent-based modeling of shopper behavior revealed that well-positioned, highly visible areas amplify pedestrian gathering effects and improve the interaction between public and commercial spaces. For instance, open dining areas within short walking distances effectively capture pedestrian flows, whereas locating visible commercial hubs at secondary nodes along major pathways strikes an optimal balance between transportation and retail functions. Consequently, the design of commercial spaces in metro stations must precisely align with pedestrian flow distribution and functional needs.

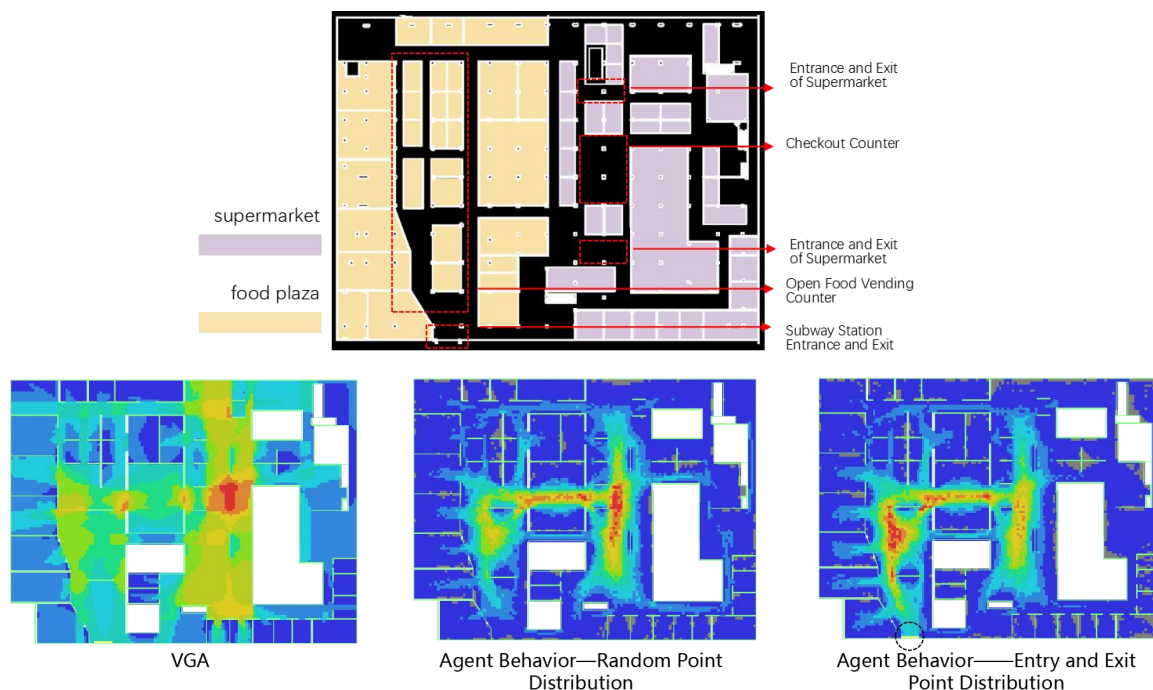


Figure 7 Underground Commercial Space (UCS)--Jiuguang Department Store for example

4. Conclusions

By integrating Space Syntax analysis with data-driven methodologies, this study presents an innovative framework for the spatial optimization of underground spaces in metro stations. The findings indicate that rationally adjusting the layout of underground public and commercial spaces can simultaneously enhance pedestrian flow efficiency, optimize user experience, and increase the economic viability of commercial areas.

Based on the case study of Jing'an Temple Metro Station, this research proposes the following optimization recommendations: First, enhance the Integration and Choice values of major corridors to improve the flow efficiency of core areas. Second, strengthen the visual design of nodes and commercial areas to create a clearer, more attractive spatial network. Third, prioritize the placement of retail and dining facilities within highly visible areas to maximize consumer dwell time and commercial revenue. From the perspective of alleviating flow congestion and enhancing user experience, efforts should also be made to improve signage systems and wayfinding frameworks within underground spaces.

This study deepens the understanding of complex behavioral patterns within underground spaces, providing theoretical insights and practical references for optimizing public underground spaces. The findings offer valuable implications for enhancing transportation efficiency, economic benefits, and social vibrancy in future high-density cities. Looking ahead, combining artificial intelligence with big data analytics could further refine the study of underground spatial behavior patterns, paving the way for more efficient and inclusive urban underground space development in the era of smart cities.

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